

Influence of surface charges on the structure of a dielectric barrier discharge in air at atmospheric pressure: experiment and modeling

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Abstract. Dielectric barrier discharges (DBD) in air at atmospheric pressure and at low frequency are mainly constituted of thin transient plasma filaments (or microdischarges) with radii of a few hundreds of micrometers. In this work, we consider a point-to-plane geometry with the dielectric covering the plane electrode. Plasma filaments are initiated by streamers, starting from the high-field region close to the point electrode. The plasma filaments deposit charges on the dielectric plate which screen the electric field and lead to an extinction of the discharge filaments. In this work we experimentally demonstrate the synchronous start of several filaments in a time range of less than a few tens of nanoseconds and we show that the charges deposited on the dielectric have a strong impact on the discharge structure. This is validated using a simple electrostatic model. Then, the dynamics of the 2D streamer propagation in the gas gap and its interaction with the dielectric plane is calculated. The influence of space charges and surface charges on the discharge structure are discussed and compared with the experiment.

PACS. 52.80.-s Electric discharges – 52.80.Hc Glow; corona – 51.50+v Electrical properties (ionization, breakdown, electron and ion mobility, etc.) – 52.65.-y Plasma simulation

1 Introduction

Currently, dielectric barrier discharges (DBDs) are widely used for various industrial applications [1,2]. At atmospheric pressure (for inter-electrode gaps between a few millimeters and a few centimeters) DBDs are generally constituted of one or several plasma filaments, also called microdischarges. In such a regime, the very short duration (a few tens of nanoseconds) and the unpredictable feature of the triggering of a filament, make them difficult to study experimentally. Guikema et al. [3] observed self-organized patterns of microdischarges (for frequency on the order of a few kHz in noble gases) controlled by the residual surface charges on dielectric surfaces. Recently, Guaitella et al. [4] described the bimodal behavior of the statistical distribution of current peaks in a cylindrical DBD, and they concluded that the high-current peak group was due to the self-triggering of several filaments (supposedly due to radiation), called collective effect (see also [5]). They also studied the impact of this effect on the injected energy in

the discharge, which directly controls the chemistry. More recently, we have shown that an auto-organization of microdischarges similar to [3] in space and time was possible in a DBD in air at atmospheric pressure and low frequency (50 Hz) [7] and demonstrated that this behavior was also controlled by phenomenon of charge deposition upon the dielectric. The influence of the charges deposited on a dielectric surface in a DBD is then essential to understand the behavior of the discharge.

In this work, we have carried out experiments and simulations to further understand the interaction of a discharge with a dielectric surface at atmospheric pressure.

2 Experimental study

2.1 Experimental set-up

The discharge is obtained in Pyrex made cell (Fig. 1). The upper cylindrical electrode (2 mm of diameter) is of tungsten, coated with a 1 mm layer of dielectric except on its tip, and linked to a 50 Hz sine high voltage power supply; the lower electrode is of salted water and grounded. We observed that in our conditions the salt concentration did not affect the discharge. The dielectric is a cylindrical Pyrex

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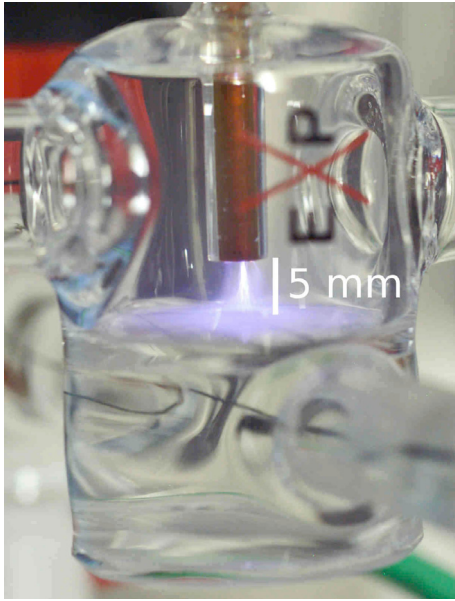


Fig. 1. (Color online) Digital camera image of the discharge.

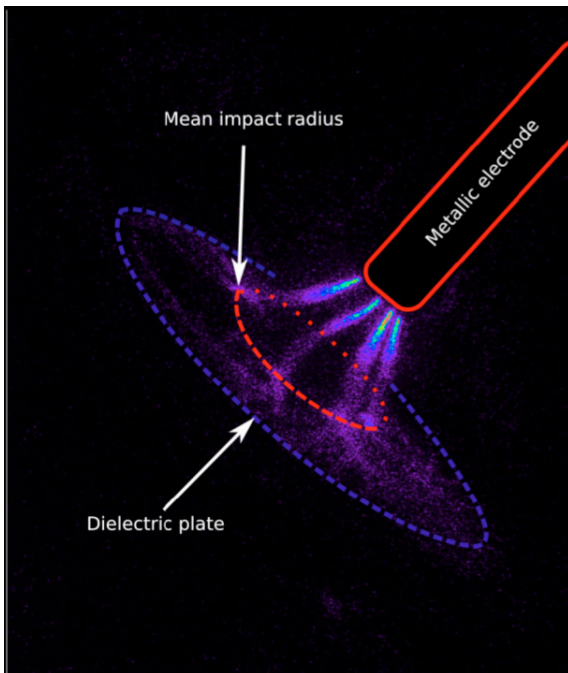


Fig. 2. (Color online) Picture of a second current peak during the positive half-cycle. The blue dashed line represents the contour of the dielectric. The red dashed line represents the mean impact radius over the dielectric of the filaments in the gaseous gap.

plate of diameter 2.2 cm and thickness 2 mm. Through the 5 mm gap flows dry air at 500 SCCM, which is released at atmospheric pressure. In these conditions, breakdown occurs for an applied voltage amplitude of 13 kV. In this study three applied voltages amplitudes are used: $V_a = 15, 18$ or 20 kV. CCD imaging is performed by an Andor iStar 734 camera, via a B7838-UV Pentax objective with a 1 ms gate and triggered by a 1 GHz Lecroy oscilloscope. Current

peaks are measured with a Fischer F33-5 Rogowski coil. A capacitor $C_m = 1.11$ nF is connected in series, for injected energy and charge transferred to be measured. Both V_a and V_m are measured with Lecroy PPE20kV probes. In this way, plasma filaments are observed using simultaneously various diagnostic methods. Images pictured by CCD are then coupled with electrical measurements [7].

2.2 Experimental results

The present work is focused on the discharges occurring when the cylindrical electrode potential is positive. In the following, we call this phase the positive half-cycle. It is interesting to note that the negative half-cycle has a quite different behavior (i.e. much more diffusive) and will not be discussed in this paper. More information on negative discharges in argon in this reactor can be found in [6]. During the positive half-cycle the number of current peaks depends on the maximum applied voltage, but remains less or equal to 4 peaks for the three voltages studied. Since the start of a streamer propagation from the anode is not predictable, and due to the time delay of electric signals in the electronic chipsets and in the wires used (~ 200 ns), it is impossible to trigger the CCD with the rising front of the current peak for picturing the discharge corresponding to the same current peak. However, it is useful to observe a picture triggered by the rising front of a current peak in order to ensure that no light is observed at least 200 ns after the rising front of the current peak. In the studied configuration, each peak (1st, 2nd, 3rd or 4th) occurs in a quite well determined time zone during the positive half-cycle. Thereby it is possible to trigger the CCD with the applied voltage and to open it during 1–2 ms such as it covers only one individual current peak knowing the rank of apparition of this current peak. We have used this approach in this work. We have noted that a first current peak in the positive half-cycle corresponds mostly to a single filament in the gaseous gap, or a few filaments above the center of the dielectric. For all cases observed, surface discharges appear from the footprint of the filament(s) forming Lichtenberg figures upon the dielectric. Figure 2 shows that the behavior of the discharges corresponding to the subsequent current peaks is different. Indeed, they are mostly constituted of several plasma filaments (about 5) which reach the dielectric plate quite far from the center of the dielectric (see Fig. 2). For these discharges, surface discharges also appear over the dielectric from the footprints of filaments. The current peaks have the same duration as a first current peak, that is ~ 50 ns, and then the streamers creating those filaments are starting in the same period of time, compared to the characteristic rising time of the power supply. Moreover, the mean impact radius defined in Figure 2 increases with the occurrence rank of the corresponding current peak. A simple explanation is possible for this phenomenon: the previous discharges have deposited charges on the dielectric plate until they shield the electric field above the charge deposits. Then the subsequent discharge filaments triggered (corresponding to the subsequent current peak) are

deviated. They eventually deposit the charges farther from the center of the dielectric [7].

In order to check whether this assumption is realistic or not, in the next section, we first use an electrostatic model to estimate surface charges on the dielectric. Second, we present the simulation of the 2D streamer propagation in the gas gap and its interaction with the dielectric plane.

3 Numerical study

3.1 Electrostatic study

We have carried out a 2D-axisymmetric electrostatic model by using COMSOL Multiphysics® in order to validate the role of the deposited charges on the dielectric on the subsequent discharges. We assumed that the negative charge deposited by the previous negative half-cycle is uniform given the diffuse feature of the negative discharge. In our experiments the very first breakdown occurred when the metallic electrode was at about 13 kV. Then we assumed that the potential difference between the metallic electrode and the water electrode for a first current peak is about 13 kV. Thereby, we obtain the Laplacian electric field in the simulation domain. We have studied the modification of the electric field by a Gaussian distribution of surface charge σ_s upon the dielectric and centered on the dielectric surface to model the effect of the charge deposition corresponding to a first current peak:

$$\sigma_s(r) = \frac{Q}{2\pi\sigma^2} \exp\left(-\frac{r^2}{\sigma^2}\right) \quad (1)$$

where σ is the characteristic length of the charge deposit (which can be chosen according to the experimental results) and Q is the total amount of charge (integration of (1) from $r = 0$ to infinity). From the measured mean impact radius of the first peaks, we can take $\sigma = 0.2$ cm. Thus we tried to determine the total charge Q which could reduce the electric field at the cylinder tip. We found that $Q = 40$ nC is the best compromise. In fact, this is the charge for which the electric field at the cylinder tip is extinguished [7]. This is a confirmation of the proposed physical explanation of experimental results, as 40 nC is precisely the average transferred charge by a first current peak in the positive half-cycle. This simple model confirms the values found experimentally such we understand them by the mechanism of the charge deposition. This model also shows that the new distribution of field in the discharge is suitable for the structure of the discharge corresponding to the second current peak observed experimentally. To better understand the interaction between the discharge and the dielectric material, as well as time evolution of surface charges, a simulation of the discharge dynamics is required. This is presented in the next section.

3.2 Simulation of the discharge dynamics

To simulate the discharge in the experimental set-up, with in particular a dielectric around the cylindrical anode, it

would be necessary to carry out three-dimensional computations. In this work, we propose to consider a simplified configuration using a hyperboloid point-to-plane geometry with a dielectric material of 1 mm thickness upon the plane cathode (located at $x = 0$). The configuration is axisymmetric with a needle anode which is assumed to be an infinite hyperboloid of revolution (tip located at $x = 0.5$ cm), and then in this work, we have carried out 2D streamer simulations. The most common and effective model to study the dynamics of streamers is based on the following drift-diffusion equations for electrons and ions coupled with Poisson's equation (e.g. [8]):

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e \mathbf{v}_e) - \nabla \cdot (D_e \nabla n_e) = S_{ph} + S_e^+ - S_e^- \quad (2)$$

$$\frac{\partial n_p}{\partial t} + \nabla \cdot (n_p \mathbf{v}_p) = S_{ph} + S_p^+ - S_p^- \quad (3)$$

$$\frac{\partial n_n}{\partial t} + \nabla \cdot (n_n \mathbf{v}_n) = S_n^+ - S_n^- \quad (4)$$

$$\nabla^2 V = -\frac{q}{\epsilon_0} (n_p - n_n - n_e) \quad (5)$$

where subscripts “e”, “p” and “n”, respectively, refer to electrons, positive and negative ions, n_i is the number density of species i , V is the potential, $\mathbf{v}_e$ is the drift velocity of electrons, D_i and μ_i are respectively the diffusion coefficient and the absolute value of mobility of species i , q is the absolute value of electron charge, and ϵ_0 is permittivity of free space. The S^+ and S^- terms stand for the rates of production and loss of charged particles. The S_{ph} term is the rate of electron-ion pair production due to photoionization in a gas volume and is calculated using the three-group SP₃ model [9]. The transport parameters and reaction rates are assumed to be explicit functions of the local reduced electric field E/N , where E is the electric field magnitude and N is the neutral density of air and are taken from [10]. The positive ion mobility is the same as in [11]. In this work, drift-diffusion equations are solved using an upwind scheme. As shown in [12], the use of this low order diffusive scheme may lead to an overestimation of the electron density and the electric field but allows to simulate the main characteristics of the streamer dynamics in air at atmospheric pressure. In this work, we have used this simple scheme to carry out a first study on the streamer/dielectric interaction. For Poisson's equation, we have used an iterative method based on iterative Strongly Implicit Procedure [13]. In this work, we use the D03EBF module of the NAG Fortran library (<http://www.nag.co.uk>). The simulation domain is 1×0.5 cm² discretized on a grid 1597×344 . In the axial direction, in the region where the streamer propagates the mesh is uniform with a cell size of $\Delta x = 5$ μ m and is refined around the dielectric interface with $\Delta x = 1$ μ m. In the radial direction, we have used a uniform grid with $\Delta r = 5$ μ m until $r = 0.15$ cm and then the grid is expanded. To calculate accurately the electric field and potential close to the hyperboloid anode in the rectilinear grid, we have applied the Ghost Fluid Method [12,14]. At the beginning of the simulation, the

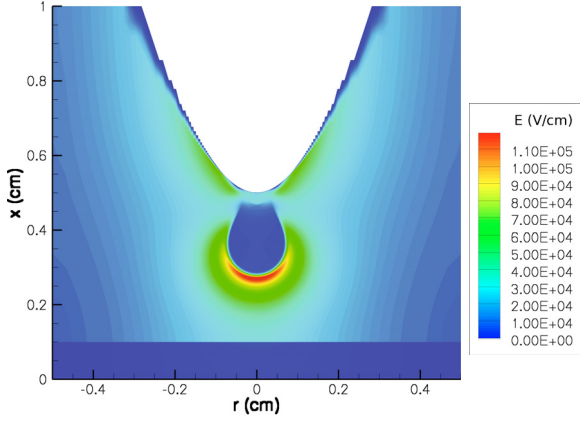


Fig. 3. (Color online) Cross-sectional view of the distribution of the magnitude of the electric field at $t = 3$ ns. The white area represents the anode. Dielectric is located between $x = 0$ and $x = 0.1$ cm.

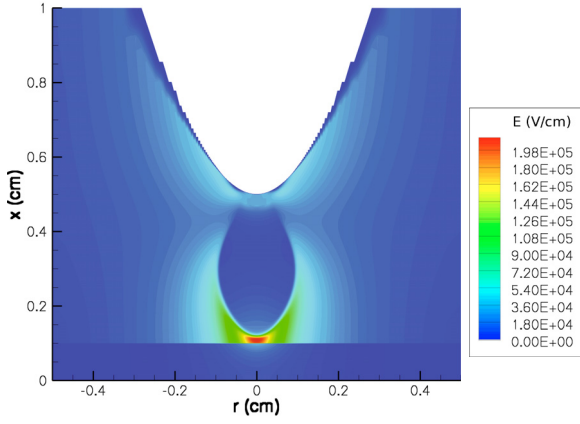


Fig. 4. (Color online) Cross-sectional view of the distribution of the magnitude of the electric field at $t = 4.5$ ns. In this simulation $\gamma = 0.1$. Dielectric is located between $x = 0$ and $x = 0.1$ cm.

positive streamer discharge is initiated by placing a neutral plasma Gaussian spot close to the anode tip:

$$n_p(x, r)|_{t=0} = n_e(x, r)|_{t=0} = n_0 \exp \left[-\frac{r^2}{\sigma_r^2} - \frac{(x - x_0)^2}{\sigma_x^2} \right] \quad (6)$$

where $x_0 = 0.5 - \sigma_x$, $\sigma_x = 0.01$ cm, and $\sigma_r = 0.01$ cm. At $t = 0$ there are no negative ions. The initial maximum density is $n_0 = 10^{14}$ cm $^{-3}$.

The propagation of the streamer from the anode to the dielectric surface lasts ~ 4.5 ns. As an example of the results obtained, Figure 3 shows the cross-sectional view of the magnitude of the electric field at $t = 3$ ns. We found out that the secondary emission via ion bombardment on the dielectric interface, introduced by a γ coefficient, plays no role during the streamer propagation in the gas gap. During this streamer propagation phase the ions and electrons close to the dielectric surface are only provided by photoionization and the increase of the surface charge density due to the ion drift is slow. During the streamer propagation, the surface charge density on the dielectric does

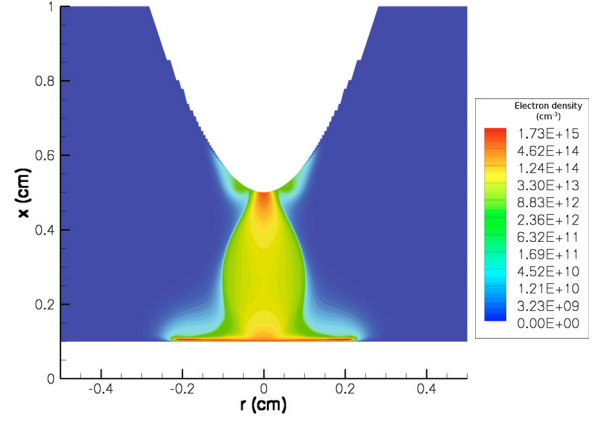


Fig. 5. (Color online) Cross-sectional view of the distribution of the electron density at $t = 8.5$ ns. White areas represent the anode and the dielectric material. In this simulation $\gamma = 0.1$.

not increase sufficiently to distort the Laplacian electric field. Figure 4 shows the cross-sectional view of the magnitude of the electric field at the moment of the impact of the streamer on the dielectric (i.e., $t = 4.5$ ns). Because of the zero boundary condition of fluxes on the dielectric (except for the secondary electron flux due to ion bombardment) a small region close to the dielectric surface of about $20 \mu\text{m}$ has a low electron concentration, and then the streamer stops when it reaches this region until the secondary processes such as the photoionization, photoelectric effect, or ion bombardment compensate for the lack of electrons. Then, a sufficient number of electrons is created to partially neutralize the streamer head. Concerning the secondary processes, some preliminary studies showed that the photoelectric effect becomes very important, or even predominant, when the streamer reaches the dielectric. In the present study, in order to save computation time we did not take into account the photoelectric effect. However, to compensate the lack of electrons and to model the streamer dynamics after its contact with the dielectric surface, we have artificially increased the γ coefficient. Note that the variation of γ has no impact on the propagation phase (from $t = 0$ to $t = 4.5$ ns). Usually, in the literature for dielectrics such as Pyrex we found $\gamma \simeq 0.01$ (e.g. [15]). In this work, we used γ as a parameter and studied the results obtained for γ varying from 0.01 to 1.

Because of the secondary electron emission from the dielectric surface the streamer head is partially neutralized when the streamer interacts with the dielectric. However, the streamer head is not fully neutralized by electron emission and the streamer eventually branches on the dielectric surface. Surface discharges are then spreading out on the dielectric. Figures 5 and 6 show cross sectional views of the electron density and of the magnitude of the electric field at $t = 8.5$ ns for $\gamma = 0.1$. Figure 6 clearly shows that during this phase the electric field is quenched except on two visible spots, which are the surface streamer heads (as the simulation is axisymmetric this discharge has a ring shape).

It is interesting to note that for all the values of the γ parameter studied in this work, the surface charge becomes significant, concerning the propagation of the surface discharge, only after 6.5 ns. That is, the surface charge produces by itself an electric field greater than E_k (the conventional breakdown electric field) only after 6.5 ns (this is checked by keeping σ_s as is for a given moment of time, while solving Poisson's equation with $\rho = 0$) i.e., after the ignition of the surface discharge. This means that the breakdown field allowing the surface discharge ignition and propagation is not exceeded because of accumulation of the surface charge, but rather because of the space charge density in a volume close the dielectric surface. This is the case even when the streamer head is easily neutralized, that is, for high γ parameters, or for a perfectly emitting dielectric surface.

As the surface discharges propagate on the dielectric material, the total surface charge increases and is 1.2 nC at $t = 9$ ns. That is 40 times less than the total surface charge for which the Laplacian electric field quenches (Sect. 3.1). This means that the screening of the electric field observed in simulations is significantly due to the space charge. To compare the different contributions of surface charge and space charge concerning the screening of the electric field, we computed the electric field due to either the surface charge or the space charge, from the results obtained at $t = 9$ ns. The results are shown in Figure 7.

We note that the electric field in the plasma channel between the anode and the dielectric given by the simulation of the discharge dynamics becomes roughly constant and on the order of the breakdown field. This means that there is no significant electron multiplication in this channel, but regular electron drift towards the dielectric. Note that this behavior was also observed for different configurations in e.g. [16,17]. Moreover, one sees that both surface and space charges have to be taken into account to understand the screening of the electric field in the considered time scales. We note that the main effect of the space charge field is the screening of the high field region close to the anode tip, while the main effect of the surface charge is the screening of the electric field close to the dielectric surface.

4 Conclusion

We have carried out an experimental study of a dielectric barrier discharge in air and at atmospheric pressure generated by a high-voltage and low frequency power supply. Generally, the effect of the surface charges deposited on the dielectric by the plasma filaments in a DBD is the screening of the electric field leading to the extinction of the discharges. In this device we found that these surface charges affect the structure of the subsequent discharges, and then correlate discharges lasting a few tens of nanoseconds separated by a few milliseconds.

Using an electrostatic simulation we demonstrated that this behavior is coherent with the total charge trans-

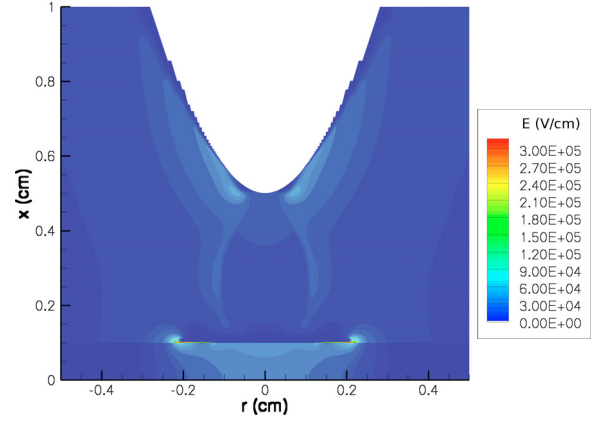


Fig. 6. (Color online) Cross-sectional view of the distribution of the magnitude of the electric field at $t = 8.5$ ns. White area represents the anode. In this simulation $\gamma = 0.1$. Dielectric is located between $x = 0$ and $x = 0.1$ cm.

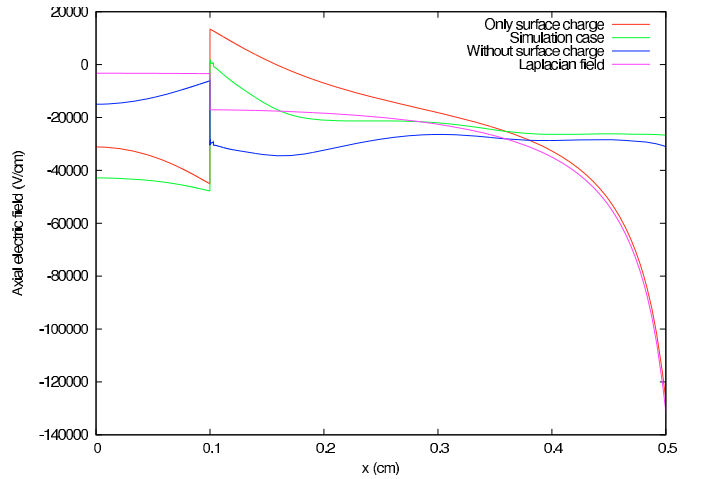


Fig. 7. (Color online) Axial electric field at time $t = 9$ ns with the different contributions of the space charge and surface charge. Dielectric is located between $x = 0$ and $x = 0.1$ cm. Red line: only the surface charge given by the streamer simulation is taken into account (no space charge). Green line: result given by the streamer simulation. Blue line: results without taking into account the surface charge. Pink thin line: Laplacian field.

ferred by the discharges corresponding to first current peaks.

Then we have simulated the discharge dynamics in a point-to-plane configuration with a dielectric upon the plane cathode. The discharge was ignited at the point anode and we have calculated its propagation in the gas gap as well as its splitting into surface discharges propagating on the dielectric material. We found that the obtained surface discharges are not ignited by an over-accumulation of charges on the dielectric surface, but rather by a residual space charge of the streamer head in a thin layer close to the dielectric. We also found that the Laplacian electric field quenches on a very short timescale (< 10 ns) because of both the contribution of the accumulated surface charge

and the residual space charge in the streamer channel. On such time scales the surface charge contributes mainly to the screening of the electric field close to the dielectric, while the space charge contributes mainly to the screening of the field close to the anode. Considering the time scales of the applied voltage at low frequency, the space charge necessarily disappears in the duration between two subsequent discharges observed experimentally. Furthermore the surface charge found in simulations for time scales of about 10 ns is not sufficient to screen the electric field by itself. Thus, we assume that the major part of the surface charge is accumulated on a much longer timescale, corresponding to the decrease of the current peak found in the measurements (~ 100 ns). To validate this assumption, it would be interesting, in a future work, to simulate the discharge dynamics on longer timescales.

In the future, it can be also interesting to simulate negative discharges, which are experimentally more diffusive than positive ones, and then to study the discharge dynamics during several cycles of the applied voltage to better understand the role of surface charges on long timescales.

Finally, we have considered simplified boundary conditions in this work, using a global γ coefficient for the dielectric. The next step is to model the surface processes in simulations by taking into account photoemission, recharging of ions, thermal influence, etc.

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